Electronics and Computer Technology

ECET 250E: Lab 05 Thevenin's Theorem and Maximum Power Transfer

Objectives

- To verify **Thevenin's Theorem**
- To determine the load resistance required for **Maximum Power Transfer**

Background

Often, we are interested in determining a node voltage or branch current of a particular sub circuit without having to determine all the other node voltages or branch currents. For instance, we may want to find the voltage and/or current for a load (and subsequently power dissipated by the load) for several different resistor values, and if the circuit is complex, calculating the value of the current or power for each value of the load resistance could be time-consuming.
There are two theorems that can be applied to situations like this: Thevenin's theorem and Norton's theorem. We do not have anything that closely resembles an ideal current source (required to implement Norton's Theorem) so we will only deal with Thevenin's theorem for this lab.

1. Thevenin's theorem

Any linear circuit can be represented at a given pair of nodes by an equivalent circuit consisting of a single voltage source in series with a single resistor.
Consider the circuit in *Figure 1*.
If we were particularly interested in the current I_L through R_L, the voltage V_L across R_L, and the power (P_L) absorbed by the load resistor R_L, then it makes sense to apply Thevenin's theorem to the circuit on the left side of terminals A and B in *Figure 1*. The Thevenin equivalent circuit is shown in *Figure 2*.
Once we have determined the Thevenin equivalent circuit (V_{th} and R_{th} in series), we need only be concerned with what is connected between the terminals A and B.
The Thevenin equivalent circuit can be determined theoretically by finding R_{th} and V_{th}.
To calculate $\mathbf{R_{th}}$, the Thevenin resistance:

1. Remove R_L from the circuit
2. Zero all independent sources in the circuit (short-circuit independent voltage sources; open-circuit ideal current sources)
3. Calculate the resistance seen looking into the circuit at terminals A and B
4. This is the Thevenin resistance R_{th}

To calculate $\mathbf{V_{th}}$, the Thevenin voltage:

1. Remove R_L from the circuit
2. Use an appropriate combination of circuit analysis methods to determine the voltage seen between the terminals A and B with all sources turned on
3. This open circuit voltage is the Thevenin voltage V_{th}

2. Maximum power transfer

Power is defined as the rate of doing work and is measured in watts. We have seen that power can be calculated from the following:

$P = IV$ $P = I^2R$ $P = V^2/R$

Every circuit has a theoretical maximum power that it can deliver to a load resistance. This power delivery will vary with the load resistance, R_L. If the load resistance is varied and the resulting power delivery is graphed against the load resistance, the resulting curve will have a maximum. This maximum will occur for $\mathbf{R_L = Rth}$.

PROCEDURE 1

1. Build the circuits of **Figures 1** on Breadboard, to complete **Table A** for $\mathbf{V_s = 12}$ **volts** and $\mathbf{V_s = 15}$ **volts.**
2. Calculate the **voltage** (V_L) across the load $\mathbf{R_L}$, the **current** (I_L) flowing through R_L and **power absorbed** by the load resistor (P_L) in circuits Figure 1. Enter these data into **Table A.** (In this procedure we won't use Thevenin Theorem, use any other theorems or laws to solve your circuit.)
3. Use all the calculated results to compare with the **measured** values in **Table A**.

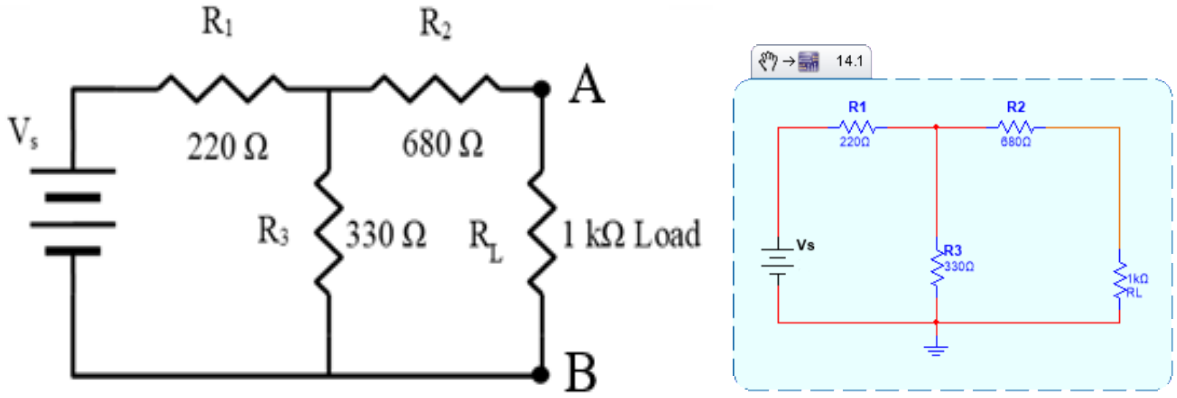


Figure 1.

TABLE A

Source voltage used ▶		12 V	15 V
I_L (mA)	Calculated		
	Measured		
V_L (V)	Calculated		
	Measured		
P_L (mW)	Calculated		
	Measured (calculate with the measured values of V,I)		

[Use the measured voltage and current values to calculate the measured power.]

PROCEDURE 2

1. Now determine the **Thevenin** equivalent circuits for **Figure 1** for both V_s **= 12 volts** and V_s **= 15 volts (Figure 2)**.
2. Build the circuit of **Figure 2** on the breadboard (use the closest available resistor in the lab room for Resistor Thevenin or series of resistors) [Keep your **Figure 1** Circuit in another side of the breadboard].
3. Use the Thevenin circuit to find the **calculated** values listed in **Table B**.
4. Use your results to compare with the **measured** values in **Table B**.

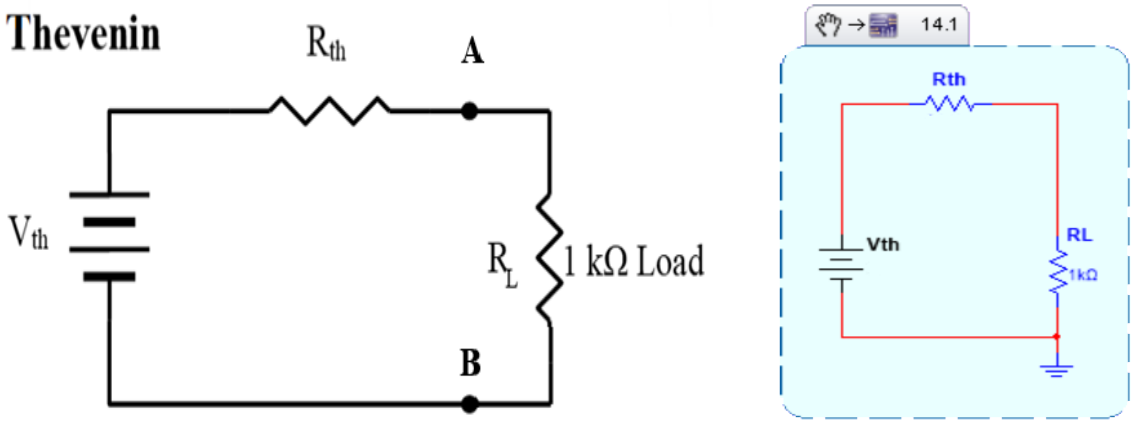


Figure 2.

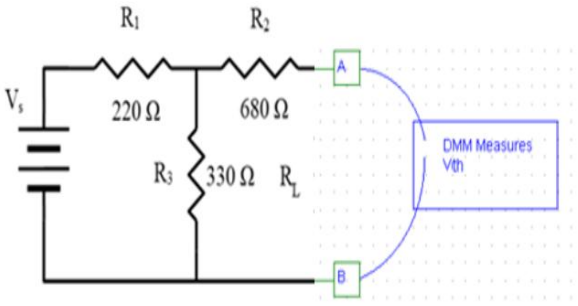
TABLE B

Source voltage used ►		**12v** (For original circuit)	**15 v** (For original circuit)
V_{th} (V)	Calculated		
	Measured		
R_{th} (Ω)	Calculated		
	Measured		
I_L (mA)	Calculated		
	Measured(From Fig. 2)		
V_L (V)	Calculated		
	Measured(From Fig. 2)		
P_L (mW)	Calculated		
	Measured (calculate with the measured values of V,I)		

How to Measure Vth and Rth Thevenin

Measuring Vth:
Build the circuit in **Figure 1** on your breadboard.

- Remove the load resistance R_L.
- Leave all sources enabled.
- Use the DMM to measure the voltage at terminals A and B, as shown in Figure below. This is **Vth**.

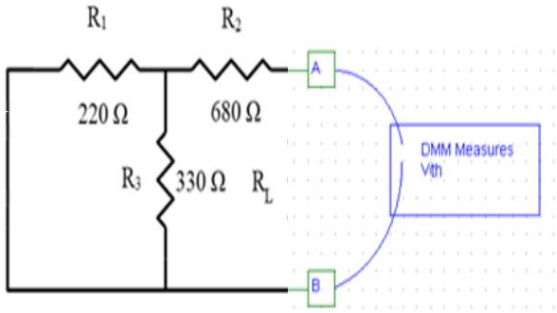


Measuring Vth from the physical circuit

Measuring Rth:

- Remove the load resistance R_L.
- Disable all sources. This means turning off all voltage sources, **physically disconnecting** them from the circuit, and providing a short circuit between network points that used to be connected to the + and - terminals of a voltage source. Make sure you do not short the supply! If there were current sources we would simply remove them from the circuit and leave the connection points open- circuited.

- Use the digital Multimeter (DMM) to measure the resistance at the terminals A and B, as shown in Figure below. Record this value of the Thevenin resistance **Rth**.



Measuring Rth from the physical circuit

1. Is the voltage across the load $\mathbf{R_L}$ (and therefore the current flowing through R_L) the same in the original series-parallel circuit and the new equivalent Thevenin circuit? Prove this with numerical evidence from your results.

2. When you measured the **open circuit voltage** in the circuit of **Figure 1**, (i.e. no load connected) how did it compare to your calculated value of $\mathbf{V_{th}}$?

3. When you measured the **circuit resistance at the output** in the circuit of **Figure 1**, (i.e. the Thevenin or internal resistance of the original series-parallel circuit) how did it compare to your calculated value of $\mathbf{R_{th}}$? What precautions must you **always** take when performing this step?

PROCEDURE 3

1. Calculate the **maximum power transfer** for **figure 2** for both $\mathbf{V_s = 12}$ volts and $\mathbf{V_s = 15}$ volts. Use your results to compare with the **measured** values in **Table C**.
2. By **varying** the load resistor $\mathbf{R_L}$ according to **Table D and E**, measure the voltage across the load resistor R_L and record them (V_L) into the related tables.
3. Calculate relevant power output and fill them into **Table D and E.**
4. **Draw** graphs (A and B) of $\mathbf{P_L}$ (measured) vs $\mathbf{R_L}$ using excel and create **maximum power transfer** curves.
5. **Label** the maximum power points on these graphs.

TABLE C

Source voltage used ►		**12 v** (For original circuit)	**15 v** (For original circuit)
P_{LMAX} **(mW)**	Calculated		
	Measured (calculate with the measured values of V)		

TABLE D (V_s = **12** volts)

R_L ohms	V_L volts	$P_L=V_L^2/R_L$ mW	V_L volts	P_L mW
Ω	Calculated	Calculated	Measured	Measured (calculate with the measured values of V)
100				
470				
812				
1200				
1800				
2700				
3300				

[For the measurements pick 820 ohms instead of 812 ohms or series of resistors.]

Graph A (V_s = **12** volts)

TABLE E (**V_s = 15** volts)

R_L ohms	V_L volts	$P_L=V_L^2/R_L$ mW	V_L volts	P_L mW
Ω	Calculated	Calculated	Measured	Measured (calculate with the measured values of V)
100				
470				
812				
1200				
1800				
2700				
3300				

[For the measurements pick 820 ohms instead of 812 ohms or series of resistors.]

Graph B (**V_s = 15** volts)

Discussion & Conclusion

a. Were the objectives met? Use your data to show that the lab objectives were met.
b. What else did you learn?
c. What problems did you encounter and how did you solve them?

[Submit the completed lab report. Attach *Pictures* of your work with all connections as part of your lab report]